TITAN_A16_AMENDMENT_001.txt
PATENT REVISION: CONTINUATION-IN-PART (CIP) INSERT
Reference: Titan A16 Original Filing
Date: January 12, 2026
Inventor: Matthew J. Campbell (Brick 1%)

SECTION 1: INSERT INTO "DETAILED DESCRIPTION"
(Paste this text after your original description of the alloy composition)

Structural Embodiment (The "Dual System" Architecture)

While the chemical composition of the Titan A16 alloy provides the fundamental material properties, the preferred embodiment of the invention utilizes a specific geometric architecture to maximize torsional damping and fatigue resistance. This architecture, referred to herein as the "Dual System" (see FIG. 1), comprises a concentric hierarchy defined by the Golden Ratio ($\varphi \approx 1.618$).

1. The Concentric Hierarchy (See FIG. 1A)
The component is not a homogeneous rod but a functionally graded system consisting of three distinct zones:
- The Core (The Anchor): A central cylinder of α+β titanium alloy (Ti-6Al-4V) serving as the primary load-bearing axis.
- The Interface (The Mediator): A surrounding layer having a radius exactly 1.618 times that of the core ($r_{interface} = r_{core} \times \varphi$). This specific thickness creates a "harmonic buffer" that prevents high-frequency shockwaves from shattering the core.
- The Outer Shell (The Mask): A surface-hardened zone (via β-Ti_3Au diffusion) that acts as the initial wear barrier.

2. The Helical Damping Geometry (See FIG. 1B)
To prevent torsional drift (rotational slippage under load), the Interface layer features a "Triple Helix" groove pattern.
- Geometry: Three continuous helical grooves offset by 120°.
- Pitch Constraint: The pitch of the helix is strictly defined by the Golden Ratio of the component's diameter (e.g., Pitch = 16.18 mm for a 10 mm core).
- Function: This geometry forces linear vibration to travel a longer, rotational path, effectively converting kinetic energy into thermal energy (heat) before it can damage the structure. The intersecting nodes of the triple helix act as "frequency traps" for resonant vibrations.

SECTION 2: INSERT INTO "BRIEF DESCRIPTION OF DRAWINGS"
(You must legally describe the image you are attaching)

FIG. 1 illustrates the Titan A16 Dual System Architecture.
FIG. 1A (Left) is a cross-sectional view showing the concentric ratio between the Core, Interface, and Outer Shell.

FIG. 1B (Right) is a longitudinal profile view showing the "Triple Helix" groove geometry and its Golden Ratio pitch alignment.

SECTION 3: REVISED CLAIMS (ADD THESE TO THE END)
(These claims protect the shape, not just the metal)

Claim [X]: An article of manufacture comprising a shaft formed from the Titan A16 alloy, wherein said shaft comprises a central core and a concentric interface layer, the ratio of the interface radius to the core radius being approximately 1.618 (φ).

Claim [Y]: The article of Claim [X], wherein the interface layer comprises a plurality of helical grooves arranged in a triple-helix configuration, said grooves having a pitch proportional to the golden ratio of the shaft diameter to induce torsional damping.

Claim [Z]: The article of Claim [Y], further comprising an outer surface layer of β-Ti_3Au (Titanium-Gold) intermetallic compound formed via diffusion hardening, providing a surface hardness exceeding 70 HRC.